From the figure,

$$\cos\theta = \frac{x}{r} \Rightarrow x = r\cos\theta$$

$$\sin\theta = \frac{y}{r} \Rightarrow y = r\sin\theta$$

The equations $x = r\cos\theta$, $y = r\sin\theta$ are called the parametric equations of the circle $x^2 + y^2 = r^2$. Here 'θ' is called the parameter and $0 \le \theta \le 2\pi$.

Example 3.26

Find the parametric equations of the circle $x^2 + y^2 = 25$

Solution

Here $r^2 = 25 \Rightarrow r = 5$

Parametric equations are $x = r\cos\theta$, $y = r\sin\theta$

$\Rightarrow x = 5\cos\theta,\ y = 5\sin\theta,\ 0 \le \theta \le 2\pi$

Exercise 3.4

1. Find the equation of the following circles having
   (i) the centre (3,5) and radius 5 units
   (ii) the centre (0,0) and radius 2 units
2. Find the centre and radius of the circle
   (i) $x^2 + y^2 = 16$
   (ii) $x^2 + y^2 - 22x - 4y + 25 = 0$
   (iii) $5x^2 + 5y^2 + 4x - 8y - 16 = 0$
   (iv) $(x+2)(x-5) + (y-2)(y-1) = 0$
3. Find the equation of the circle whose centre is (–3, –2) and having circumference 16π
4. Find the equation of the circle whose centre is (2,3) and which passes through (1,4)
5. Find the equation of the circle passing through the points (0, 1),(4, 3) and (1, –1).
6. Find the equation of the circle on the line joining the points (1,0), (0,1) and having its centre on the line $x + y = 1$
7. If the lines $x + y = 6$ and $x + 2y = 4$ are diameters of the circle, and the circle passes through the point (2, 6) then find its equation.
8. Find the equation of the circle having (4, 7) and (–2, 5) as the extremities of a diameter.
9. Find the Cartesian equation of the circle whose parametric equations are $x = 3\cos\theta, y = 3\sin\theta,\ 0 \le \theta \le 2\pi$.

3.4.5 Tangents

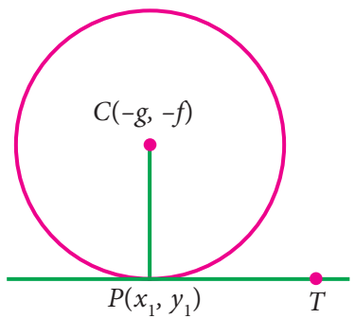


Fig. 3.7

The equation of the tangent to the circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

Corollary:

The equation of the tangent at (x_1, y_1) to the circle

$x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$

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